

add element to set

```
a={1,2,3}  
a.add(4)  
print(a)
```

```
{1, 2, 3, 4}
```

remove element from set

```
a={1,2,3,4}  
a.remove(4)  
print(a)
```

```
{1, 2, 3}
```

```
a={1,2,3,4}  
a.discard(4)  
print(a)
```

```
{1, 2, 3}
```

union of two sets

```
a={1,2,3,4}  
b={1,2,4}  
r=a.union(b)  
print(r)
```

```
{1, 2, 3, 4}
```

intersection of two sets

```
a={1,2,3}  
b={1,2,4}  
r=a.intersection(b)  
print(r)
```

```
{1, 2}
```

```
a={1,2,3}
b={1,2,4}
r=a&b
print(r)
```

```
{1, 2}
```

difference of sets

```
a={1,2,3}
b={1,2}
r=a.difference(b)
print(r)
```

```
{3}
```

check subset

```
a={1,2}
b={1,2,3}
r=a.issubset(b)
print(r)
```

```
True
```

set length finding

```
a={1,2,3}
print(len(a))
```

```
3
```

clear a set

```
a={1,2,3}
a.clear()
print(a)
```

```
set()
```

symettric difference

```
a={1,2,3,4}
b={2,3}
```

```
r=a.symmetric_difference(b)
print(r)
```

```
{1, 4}
```

```
a={1,2,3,4}
b={2,3}
print(a^b)
```

```
{1, 4}
```

convert list to set

```
a=[1,2,2,3]
print(set(a))
```

```
{1, 2, 3}
```

creating dict from two lists

```
a=['a','b','c']
b=[1,2,3]
r=dict(zip(a,b))
print(r)
```

```
{'a': 1, 'b': 2, 'c': 3}
```

update dict values

```
a={"a":1,"b":2}
a["a"]=3
print(a)
a.update({"c":4})
print(a)
```

```
{'a': 3, 'b': 2}
```

```
{'a': 3, 'b': 2, 'c': 4}
```

remove key from dict

```
a={'a':1,'b':2,'c':3}
a.pop("a")
print(a)
del a['b']
print(a)
```

```
{'b': 2, 'c': 3}
{'c': 3}
```

Check Key Existence

```
s={'a':1,'b':2,'c':3}
print('a' in s)
```

True

Iterate Over Dictionary

```
s={'a':1,'b':2}
for i,j in s.items():
    print(i,j,end=' ')
```

a 1 b 2

Dictionary Length

```
a={"a":1,"b":2}
print(len(a))
```

2

17. Merge Two Dictionaries

```
a={"a":1}
b={"b":2}
a.update(b)
print(a)
```

{'a': 1, 'b': 2}

18. Get Value with Default

```
a={"a":1}
r=a.get("b")
print(r)
```

None

19. Count Frequency of Elements

```
def freq(lst):  
    res={}  
    for num in lst:  
        res.update({num:lst.count(num)})  
    return res  
print(freq([1,2,2,3]))
```

```
{1: 1, 2: 2, 3: 1}
```

20. Invert a Dictionary

```
s={"a":1,"b":2}  
flip={}  
for k,v in s.items():  
    flip[v]=k  
print(flip)
```

```
{1: 'a', 2: 'b'}
```

21. Find Key with Maximum Value

```
s={"a":12,"b":89,"c":9}  
res=max(s, key=s.get)  
print(res)
```

```
b
```

22. Sort Dictionary by Values

```
d = {"a": 3, "b": 1, "c": 2}  
res = sorted(d.items(), key=lambda x: x[1])  
print(res)
```

```
[('a', 3), ('b', 1), ('c', 2)],)
```

23. Create Dictionary of Squares

```
sq={}  
for i in range(1,4):  
    sq[i]=i*i  
print(sq)
```

```
{1: 1, 2: 4, 3: 9}
```

24. Filter Dictionary by Value Condition

```
d={"a": 10, "b": 5, "c": 15}
res={key:value for key,value in d.items() if value>10}
print(res)
```

```
{'c': 15}
```

25. Combine Values of Duplicate Keys

```
d1 = {"a": 1, "b": 2}
d2 = {"a": 3, "c": 4}
res = d1.copy()
for key in d2:
    if key in res:
        res[key] = res[key] + d2[key]
    else:
        res[key] = d2[key]
print(res)
```

```
{'a': 4, 'b': 2, 'c': 4}
```

26. Count Word Frequency in Sentence

```
n="apple banana apple"
sen=n.split()
d={}
for i in sen:
    if i in d:
        d[i]+=1
    else:
        d[i]=1
print(d)
```

```
{'apple': 2, 'banana': 1}
```

27. Remove Duplicate Values from Dictionary

```
n = {"a": 1, "b": 2, "c": 1}
res = {}
for key, value in n.items():
    if value not in res.values():
        res[key] = value
print(res)
```

```
{'a': 1, 'b': 2}
```

28. Find Common Keys in Two Dictionaries

```
d1={"a":1,"b":2}
d2={"b":3,"c":4}
res=[]
for key in d1:
    if key in d2:
        res.append(key)
print(res)
```

```
['b']
```

29. Swap Keys and Values Safely

```
n={"x":1,"y":2}
res={v:k for k,v in n.items()}
print(res)
```

```
{1: 'x', 2: 'y'}
```

30. Delete Items by Value

```
n = {"a": 1, "b": 2, "c": 1}
res = {}
for key, value in n.items():
    if value != 1:
        res[key] = value
print(res)
```

```
{'b': 2}
```

